

UNIT 1

1. Introduction to Natural Language Processing (NLP)

What is NLP?

Natural Language Processing is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human language (text or speech).

Why NLP is needed

Humans communicate using natural language

Computers understand only binary/mathematical form

NLP acts as a bridge between human language and machines

Applications of NLP

Machine Translation (Google Translate)

Speech Recognition (Alexa, Siri)

Chatbots

Sentiment Analysis

Information Retrieval

Text Summarization

Levels of NLP

Lexical Analysis – word level

Syntactic Analysis – grammar

Semantic Analysis – meaning

Discourse Analysis – context

Pragmatic Analysis – real-world meaning

Challenges in NLP

Ambiguity (word has multiple meanings)

Synonyms

Context understanding

Language diversity

📌 Exam Tip: Write definition + applications + challenges.

2. Words

What is a Word in NLP?

A word is the basic unit of language used for processing text.

Word-related Concepts

Token: Individual word after tokenization

Example: "I love NLP" → [I, love, NLP]

Vocabulary: Set of unique words

Type vs Token

Type → unique word

Token → total occurrences

Word Normalization

Lowercasing

Stemming (playing → play)

Lemmatization (better → good)

Out-of-Vocabulary (OOV) Words

Words not present in training data.

📌 Exam Tip: Mention tokenization + normalization.

3. Regular Expressions and Automata

Regular Expressions (Regex)

What is Regex?

A pattern matching language used to identify strings in text.

Common Regex Symbols

Symbol Meaning

. Any character

* Zero or more

+ One or more

? Optional

[] Character set

Uses of Regex in NLP

Tokenization

Removing punctuation

Identifying dates, emails, phone numbers

Automata

Finite State Automaton (FSA)

A mathematical model with:

States

Transitions

Start state

Final state

Types

DFA (Deterministic)

NFA (Non-Deterministic)

Use in NLP

Spell checking

Morphological analysis

Token recognition

📌 Exam Tip: Draw simple FSA diagram if possible.

4. Words and Transducers

What is a Transducer?

A Finite State Transducer (FST) maps input strings to output strings.

Difference between FSA and FST

FSA	FST
Accepts or rejects	Transforms input
Recognition	Conversion

Uses of FST :

Morphological analysis

Example:

Play + ed → played

Pronunciation mapping

📌 Exam Tip: Give example of word transformation.

5. N-grams

What is an N-gram?

A sequence of N consecutive words from text.

Types

Unigram (N=1): I

Bigram (N=2): I love

Trigram (N=3): I love NLP

Why N-grams are used

Language modeling

Predict next word

Spell checking

Speech recognition

N-gram Probability

$$P(w_n | w_{n-1}) = \frac{\text{Count}(w_{n-1}, w_n)}{\text{Count}(w_{n-1})}$$

Problems

Data sparsity

Zero probability

📌 Exam Tip: Write definition + example + formula.

6. **Part-of-Speech (POS) Tagging**

What is POS Tagging?

Assigning grammatical labels (noun, verb, adjective) to words.

Example

Sentence:

“I eat apples”

Word	POS
I	Pronoun
Eat	Verb
Apples	Noun

Approaches

Rule-based

Statistical

Machine Learning

Applications

Parsing

Machine translation

Information extraction

📌 Exam Tip: Give sentence example with tags.

7. Hidden Markov Model (HMM)

What is HMM?

A statistical model where:

States are hidden

Observations are visible

HMM Components

States (POS tags)

Observations (words)

Transition Probability

Emission Probability

Example

States → Noun, Verb

Observations → words

Use of HMM

POS tagging

Speech recognition

Key Algorithms

Viterbi Algorithm (best tag sequence)

Forward-Backward

📌 Exam Tip: Draw HMM diagram + mention Viterbi.

8. Maximum Entropy (MaxEnt) Model

What is Maximum Entropy?

A probabilistic model that selects the most uniform (maximum entropy) distribution satisfying given constraints.

Main Idea

Uses features

Makes no assumptions beyond constraints

Advantages

Can use many features

No independence assumption

Applications

POS tagging

Text classification

Comparison with HMM

HMM.	MaxEnt
Generative.	Discriminative
Strong assumptions.	Flexible
Less features.	Many features

📌 Exam Tip: Write advantages + comparison.

UNIT 1 – QUICK REVISION TABLE

Topic	Key Focus
NLP Intro.	Definition, applications
Words.	Token, normalization
Regex & Automata	Pattern matching
Transducers	Word transformation

N-grams

Language model

POS Tagging

Grammar labeling

HMM

Probabilistic tagging

MaxEnt

Feature-based model